

Lift & Drag Constants — 8045 Propeller

A2212 1000KV, 3S • QuadFS • derived from UIUC wind-tunnel measurements

1. The Constants

$$T = k_f \cdot \omega^2 \quad Q = k_m \cdot \omega^2 \quad (\omega \text{ in rad/s})$$

Constant	rad/s units	RPM units
k_f (lift / thrust)	$5.79 \times 10^{-6} \text{ N}/(\text{rad/s})^2$	$6.35 \times 10^{-8} \text{ N}/\text{RPM}^2$
k_m (drag / torque)	$7.68 \times 10^{-8} \text{ N}\cdot\text{m}/(\text{rad/s})^2$	$8.43 \times 10^{-10} \text{ N}\cdot\text{m}/\text{RPM}^2$

Unit rule: pick ONE unit system and use it everywhere. The rad/s and RPM values differ by $(\pi/30)^2 \approx 1/91$; mixing them is invisible in code review and catastrophic in the mixer. Encode the unit in the name, e.g. `KF_N_PER_RADPS2`.

2. The Useful Ratios

Ratio	Value	Units	Where it is used
k_f / k_m	75.4	m^{-1}	yaw vs thrust channel weight in the mixer inverse
k_m / k_f	13.3	mm	reaction torque per Newton of thrust (yaw mixer gain)
$k_f \cdot L / k_m$ (F330, $L = 0.165 \text{ m}$)	12.4	—	roll/pitch vs yaw authority: same motor-speed differential gives ~12× more roll torque than yaw torque
$k_f \cdot L / k_m$ (F450, $L = 0.225 \text{ m}$)	16.9	—	same, F450 arms

Note k_f/k_m is NOT dimensionless: N per N·m leaves 1/length. The dimensionless authority ratio $k_f \cdot L / k_m$ is the quantitative statement of why yaw is the authority-starved axis on this propeller.

3. Data Source

- Propeller (thrust & torque vs RPM, measured): UIUC Propeller Database Vol. 1 — GWS Slow Flyer 8x4.3 static test, the geometry the generic 8045 is cloned from. Load cell + torque cell.

https://m-selig.ae.illinois.edu/props/volume-1/data/gwssf_8x4.3_static_2869rd.txt

- Motor (achievable RPM): A2212/13T manufacturer table — GWS 8x4 prop reaches 9,660 RPM at 10.9 V (charged 3S), 6.5 A.

https://www.rhydolabz.com/documents/26/BLDC_A2212_13T.pdf

Citation: Brandt, J.B. & Selig, M.S., "Propeller Performance Data at Low Reynolds Numbers," AIAA 2011-1255.

4. Derivation (compact)

The file gives 16 rows of RPM, CT, CP. Averages: $CT = 0.1094$, $CP = 0.0449$ — flat across the whole RPM sweep, which is the experimental proof that $T \propto \omega^2$ and $Q \propto \omega^2$ hold.

Standard definitions (n in rev/s, $\rho = 1.225 \text{ kg/m}^3$, $D = 0.2032 \text{ m}$):

$$T = CT \rho n^2 D^4 \quad P = CP \rho n^3 D^5 \quad Q = P / 2\pi n = (CP/2\pi) \rho n^2 D^5$$

Substitute $n = \omega/2\pi$:

$$k_f = C_T \rho D^4 / 4\pi^2 = 5.79 \times 10^{-6} \quad k_m = C_P \rho D^5 / 8\pi^3 = 7.68 \times 10^{-8}$$

($8\pi^3$ = the squared unit conversion $4\pi^2$ times the power→torque division 2π)

5. Operating Points (per motor)

Point	RPM	ω (rad/s)	Thrust	Torque
Hover, 800 g quad	5,560	582	1.96 N (200 g)	25.9 mN·m
End of measured data	7,000	733	3.11 N (317 g)	41.3 mN·m
Max, charged 3S	9,660	1,012	5.93 N (604 g)	78.6 mN·m
Max, mid-flight (~10.4 V)	~9,200	963	~5.4 N (~550 g)	~71 mN·m

Cross-validation: the hover point matches the aircraft's observed behavior, and k_f applied to the manufacturer's 9,660 RPM gives 604 g — inside the 550–620 g range reported by independent thrust-stand tests. Two measurement chains agree.

6. Caveats

- The tested prop is the GWS SF 8x4.3; the ABS 8045 is a clone (pitch 4.5). Hold k_f to ± 10 %, k_m to ± 15 %.
- Static data: in forward flight, thrust at a given RPM drops ~10–15 % from inflow.
- These are lumped rotor constants, not airfoil CL/CD. They enter the dynamics as total thrust = $k_f \cdot \sum \omega_i^2$ and yaw torque = $k_m \cdot (\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$.
- To measure your exact prop: motor on a tared scale, RPM from ESC telemetry, five throttle points, fit T vs ω^2 ; scale k_m by the same correction found for k_f .